



EXECUTIVE BRIEFING • ANALYTICS EXPERIENCE

From Food Waste Data to Operational Intelligence

PlateSight Analytics Tab: Turning automated waste records into operational visibility, financial understanding, and actionable interventions.

PREPARED FOR

Dolphin Group of Hotels

Ramoji Film City, Hyderabad
(Sitara • Sahara • Tara • Eureka)

SCOPE & FOCUS

Analytics Experience

Centralized Web Management Portal
ai.platesight.in/analytics

PREPARED BY & DATE

PlateSight Intelligence

Gandhaar Joshi, Co-Founder
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One Analytical Workflow Can Bring Multiple Sources Together

PlateSight consolidates automated hardware logging, existing kitchen spreadsheets, and manual audits into a single pipeline.

THREE DATA PILLARS

Multi-Stream Data Harmonization

01 • AI Vision & Load-Cell Stations

Edge-computed plate and tray logs provide instant food segmentation and certified kilogram weights without manual kitchen logging.

02 • Kitchen Excel Sheets & ERP

Automated ingestion of daily kitchen production, pickup, and wastage reports (e.g. Sahara & Sitara daily logs) to establish historical baselines.

03 • Managerial & Audit Overrides

Authorized supervisor portal to input guaranteed banquet Pax, menu metadata, and food-safety disposition flags.

THE CENTRAL DATA PIPELINE

From Raw Logs to Executive Action

Step 1: Standardization & Unit Normalization

Normalizes diverse kitchen units (kg, portions, pans) and maps raw food labels to standardized hotel master recipes.

Step 2: Relational Aggregation

Organizes records by property (Sitara, Sahara), meal shift (Lunch, Dinner), and service format (Buffet, Banquet).

Step 3: Rules Engine & Variance Triggers

Compares prepared vs consumed vs discarded against historical benchmarks to detect waste anomalies.

Step 4: Executive Dashboards & Action Cards

Surfaces instant financial valuation and actionable batch-size adjustment recommendations.

Data Governance: Historical spreadsheets bootstrap intelligence immediately, ensuring baseline visibility prior to physical station rollout.

Start With the Full Food-Waste Picture

Top-level KPI cards establish executive visibility across total preparation, consumption, waste, cost, and guest volume.

EXECUTIVE FOOD INTELLIGENCE & YIELD OPTIMIZATION

Culinary Analytics & Waste Intelligence

Transforming raw kitchen production, banquet buffet returns, and unconsumed food records into actionable operational decisions.

Upload Report (.xlsx)

Current interface snapshot: Top-level operational KPI cards (ai.platesight.in/analytics) • Values shown represent current platform demonstration data.

VOLUME & INTAKE

Prepared vs. Consumed

- Total Food Prepared: Establishes total kitchen production volume entering service.
- Consumption Rate: Measures true guest intake (e.g. 74.8% intake, 546g/guest).
- Waste % & g/PAX: Normalizes discard volume against changing banquet group sizes.

COST & BUFFER DRIFT

Financial Valuation & Drift

- Total Waste Cost: Maps discard kilograms directly to hotel ingredient cost sheets.
- Waste Cost / Guest: Isolates per-cover margin erosion for banquet pricing review.
- Prepared vs Planned: Quantifies kitchen buffer inflation against contracted pax.

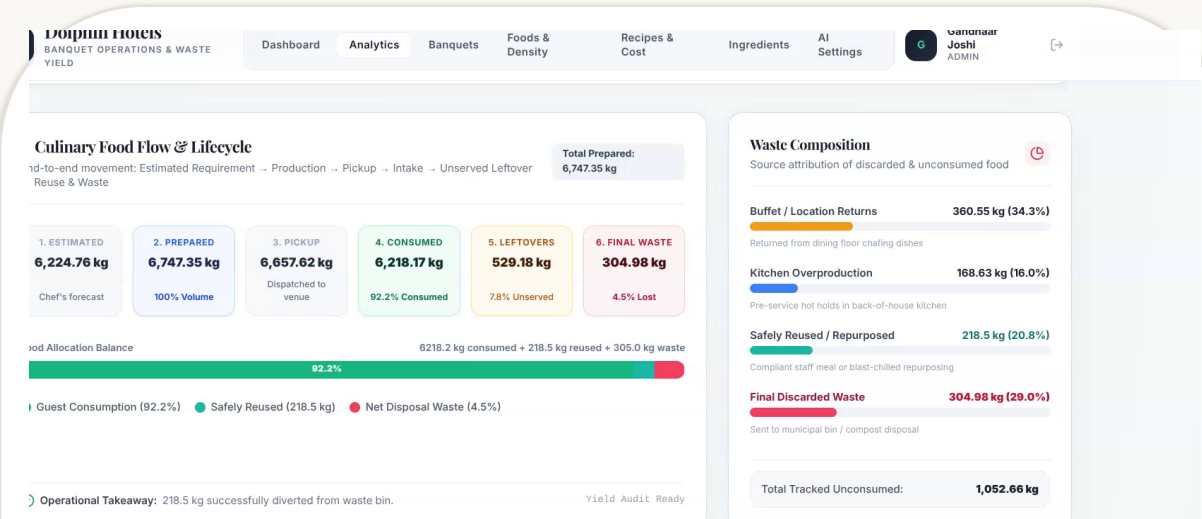
RESOURCE RECOVERY

Leftover Segregation

- Kitchen vs Buffet Split: Differentiates unserved pan holding from buffet table returns.
- Diverted / Reused Food: Tracks safe culinary repurposing under hotel food-safety SOPs.
- Clean Separation: Strictly avoids counting clean leftovers as unrecoverable waste.

Understand Where Food Goes Before Calling It Waste

Tracking food through distinct operational stages prevents misattributing kitchen holding, buffet returns, or safe reuse as landfill waste.



Current interface snapshot: Culinary Food Flow & Waste Composition panel.

CULINARY MASS BALANCE

- Mass-Balance Ledger: Total Prepared = Consumed + Cold-Chain Leftovers + Final Discard.
- Cold-Chain Preservation: Safe unserved holding is tracked separately under HACCP temp logs.
- Culinary Credibility: Sous chefs audit disposition before final service sign-off.

LIFECYCLE MASS-BALANCE

End-to-End Kitchen Journey

- Estimated Requirement: Forecasted volume based on banquet PAX & per-head specs.
- Kitchen Production: Actual cooked food output dispatched from central hot kitchens.
- Service Intake: Food picked up and consumed by banquet and restaurant guests.
- Surplus Disposition: Separation into unserved leftovers vs plate scrapings.

WASTE COMPOSITION

Crucial Operational Categorization

- Buffet Returns: Unserved hot/cold chafing dishes under temperature control.
- Safely Reused: Rapid-chilled items repurposed under hotel food-safety SOPs.
- Final Discard: Contaminated plate scrapings and post-service spoilage.
- **Rigorous Rule: Reused food is strictly separated and NEVER double-counted as waste.**

A Signal Is the Start of an Investigation

Rules-driven intelligence translates operational anomalies into targeted kitchen inquiries, not unverified root-cause accusations.

Culinary operations recorded 6747.4 kg prepared and 305.0 kg final waste (4.52% waste rate) across 10 dining shifts. Direct financial loss stands at ₹23,194 (3.3 ₹/guest). Dinner shift generated peak loss, with top 3 dishes accounting for 109.9 kg of total discards.

 HIGH-LOSS RECIPE

CRITICAL

Plain Rice Accounting for Disproportionate Waste

Observation: Plain Rice is currently the #1 discarded menu item with 55.4 kg waste (5.4% waste rate).

Data Metric: Financial loss of \$1,108 across 2 meal preparations.

 RECOMMENDED ACTION

Calibrate standard batch portion sizes for Plain Rice by approximately 10–15% based on actual intake.

Current interface snapshot: Operational Intelligence & Action Plan cards with priority filters.

LIVE ACTION CARD SAMPLES

Direct Recommendations Generated in Dashboard

- [CRITICAL] Plain Rice Batch Size: Cut dinner buffet primary batch by 25% (Sahara Kitchen). Estimated impact: -14 kg/day.
- [ATTENTION] Sambar Replenishment: Switch to 2-stage half-pan holding in final 45 mins. Estimated impact: -8 kg/day.
- [PERFORMING WELL] Bread & Bakery Station: Waste index < 3.2% across last 7 consecutive services.

THE 5-STEP ACTION FRAMEWORK

How Observations Become Results

- 1. Metric Signal: Identifies anomaly (e.g. Dinner buffet waste >30%).
- 2. Data Indication: Isolates specific recipe drift (e.g. Plain Rice over-batching).
- 3. Investigation: Sous chef inspects pan replenishment timing & headcounts.
- 4. Kitchen Action: Implement staged batch cooking (50% base + 25% + 25%).
- 5. Verification: Measure waste reduction across next 5 comparable dinner services.

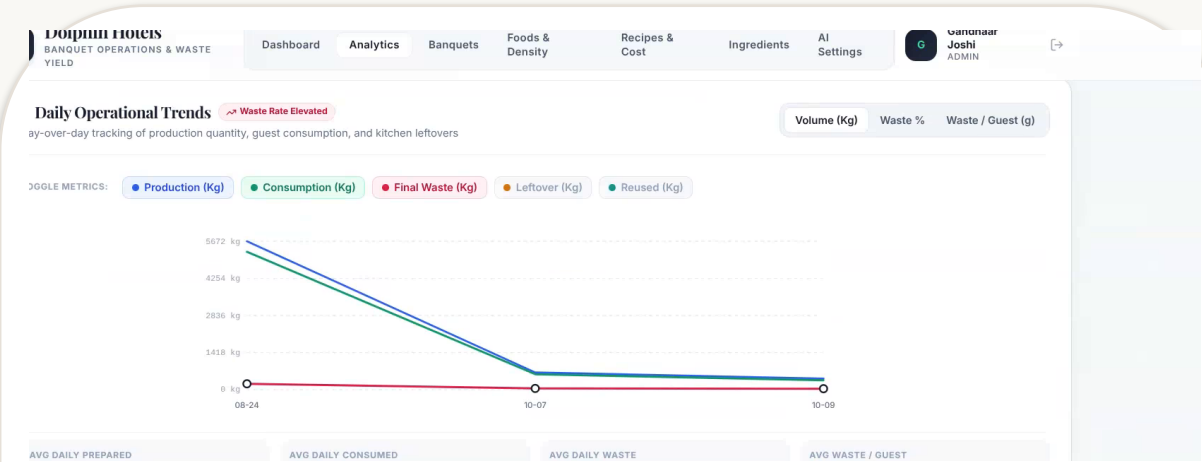
TRIAGE & TRANSPARENCY

Rules-Driven, Not Black-Box

- **Priority Triage:** Filter by Critical, Attention, Performing Well, and Information.
- **Deterministic Logic:** Actions are based on hotel-configured rules, not hallucinated AI claims.

Separate Total Volume From Performance Rates

Day-over-day tracking reveals whether rising waste stems from operational inefficiency or simply higher guest volume.



Current interface snapshot: Daily Operational Trends (Volume, Waste %, Waste/Guest).

NORMALIZED BENCHMARKING

Grams-Per-Guest (g/PAX) Target Metrics

- Gross Volume vs True Rate: 100kg waste with 2,000 PAX (50g/guest) is far more efficient than 30kg waste with 100 PAX (300g/guest).
- Target Hospitality Benchmark: <80g/PAX for multi-dish banquet buffets; <45g/PAX for business luncheons.

DUAL-AXIS EVALUATION

Why Normalization Matters

- Gross Kilograms (kg): Indicates total handling volume and physical disposal load.
- Waste Percentage (%): Measures kitchen batch accuracy relative to total cooked volume.
- Grams per Guest (g/PAX): The gold standard hospitality benchmark. Evaluates whether 50kg waste came from 200 guests (crisis) or 2,000 guests (highly efficient).

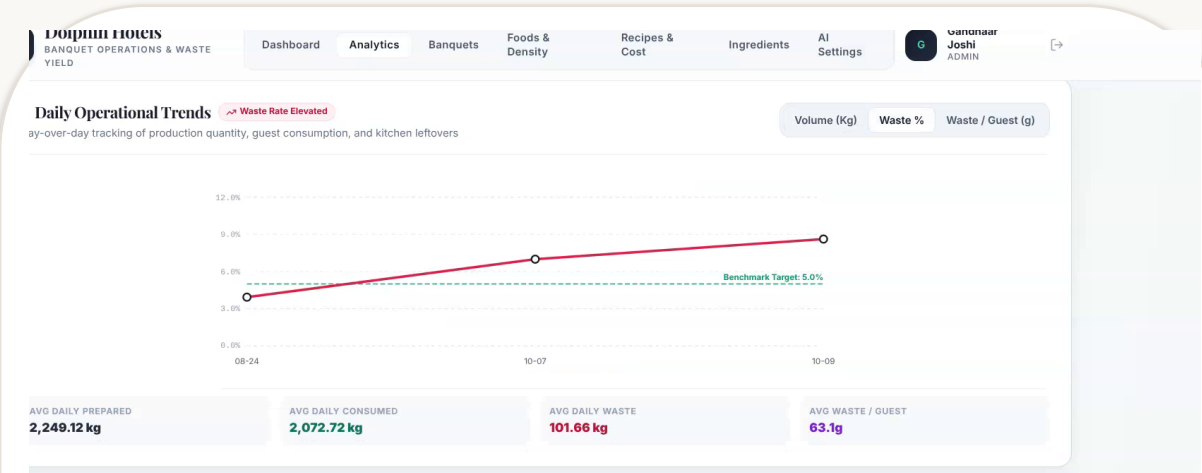
MULTI-STREAM TOGGLES

Tracking Cross-Stream Dynamics

- Metric Toggles: Seamlessly switch views between Production, Consumption, Final Waste, Leftover, and Reused.
- Management Insight: Verifies whether waste reduction is driven by leaner cooking batches or improved leftover recovery.
- Rolling Averages: Computes daily average production and grams per guest to smooth out weekend banquet fluctuations.

Find Out Which Operating Context Needs Attention

Grouping performance by meal shift and service format pinpoints structural operational differences across banquet and outlet operations.



Current interface snapshot: Session-wise Breakdown & Service Type Operational Yield.

OPERATIONAL SHIFT AUDIT

Key Session Yield Observations

- Dinner Shift Disproportion: Dinner accounts for over 58% of aggregate daily waste due to late-buffet presentation requirements.
- Buffet Pan Tapering SOP: Implementing half-pan replenishment in the final 45 minutes saves ~15–20% without impacting guest satisfaction.

SHIFT PROFILES

Session-Wise Dynamics

- Breakfast: Standardized pickups, high predictability, low waste intensity.
- Lunch: High turnover, compact service window, moderate buffet recovery.
- **Dinner: Consistently highest waste intensity due to prolonged service holding and late-buffet presentation requirements.**
- Snacks / Hi-Tea: Variable pickup; sensitive to precise conference timing.

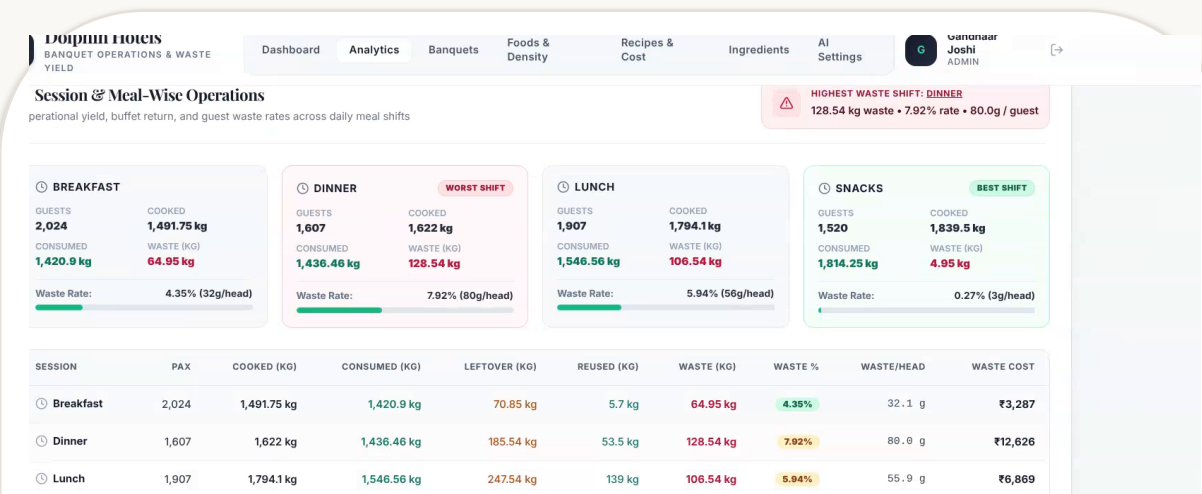
SERVICE YIELD

Buffet vs Banquet vs À la Carte

- Buffets: Suffer from 'full pan' visual presentation pressure until closing.
- Banquets: High volume, fixed menus, contractual over-buffering against guarantees.
- À la Carte: On-demand preparation delivers tightest margins and lowest waste (~3–5%).
- **Targeted Action: Interventions focus specifically where 85%+ of waste occurs: Buffets & Banquets.**

Review One Event, Then Compare Like-for-Like Events

Granular function records enable event-by-event post-mortems while building historical benchmarks across event categories.



Current interface snapshot: Banquet Event Performance Table with configurable thresholds.

FUNCTION AUDIT CRITERIA

Automated BEO Threshold Triggers

- Guaranteed vs Actual Variance: Instantly flags events where guest turnout deviated by >15% from contract guarantee.
- Margin Protection: Prevents unfair attribution of client no-shows to kitchen overproduction.

EVENT AUDIT MATRIX

Comprehensive Function Auditing

- Event Metadata: Function Name, Hall (Sitara, Sahara), Date, and Guaranteed PAX.
- Volume Reconciliation: Estimated Requirement vs Actual Cooked vs Consumed kilograms.
- Disposition Tracking: Leftovers recovered, safely reused, and final discarded kilograms.
- Financial Impact: Total Waste Cost (₹) and Cost per PAX.
- Compliance Status: Automated status pill (Compliant, Review, Critical) via configurable thresholds.

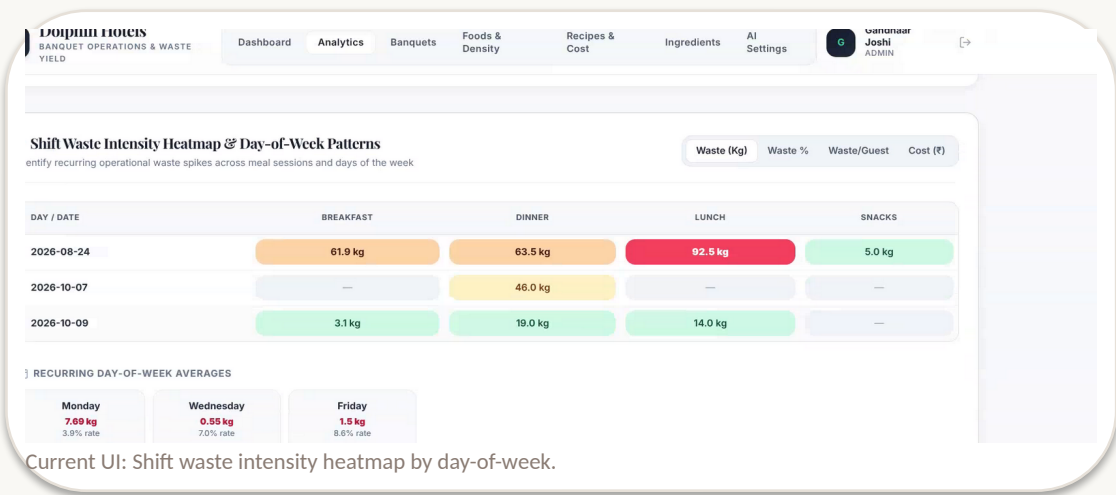
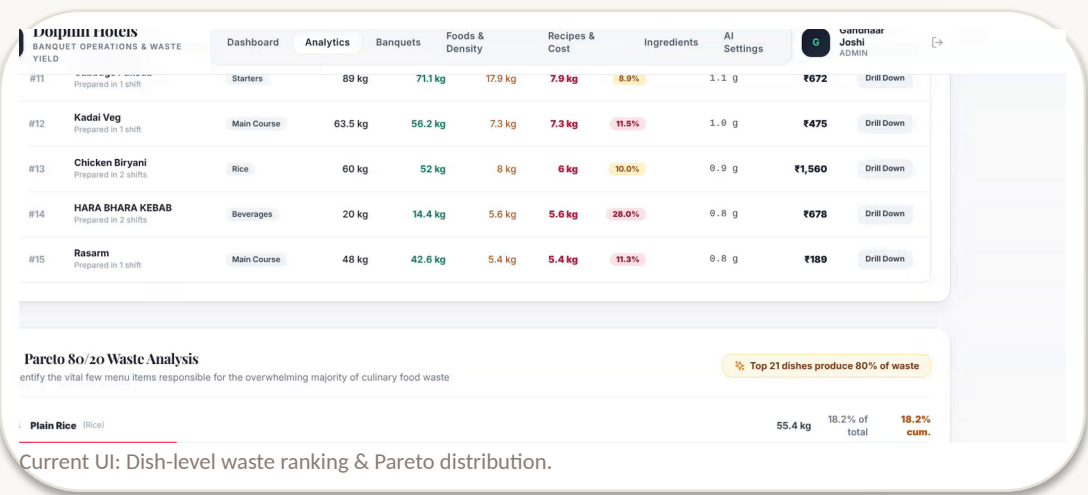
EVENT BENCHMARKING

Like-for-Like Event Analysis

- Event Type Segmentation: Compare lavish wedding banquets against corporate seminars of identical PAX.
- Menu Complexity Impact: Evaluate whether 10-course buffets generate 40% more waste than 6-course formats.
- Sales Feedback Loop: Feeds actual PAX eating patterns directly back into banquet sales contracts.

Move From Total Waste to the Specific Dishes and Periods Behind It

Combining item-level Pareto rankings with temporal heatmaps directs culinary focus to the high-impact recipes and recurring shift windows.



PARETO DISTRIBUTION

Top-Waste Dish Concentration

- The 80/20 Rule: In Indian hotel banquets, 4-5 core staple recipes consistently generate over 70% of total discarded weight (Plain Rice, Sambar, Rasam, Dal, Biryani).
- Actionable Priority: Eliminating overproduction on just these 4 items solves the vast majority of volume and disposal expense.

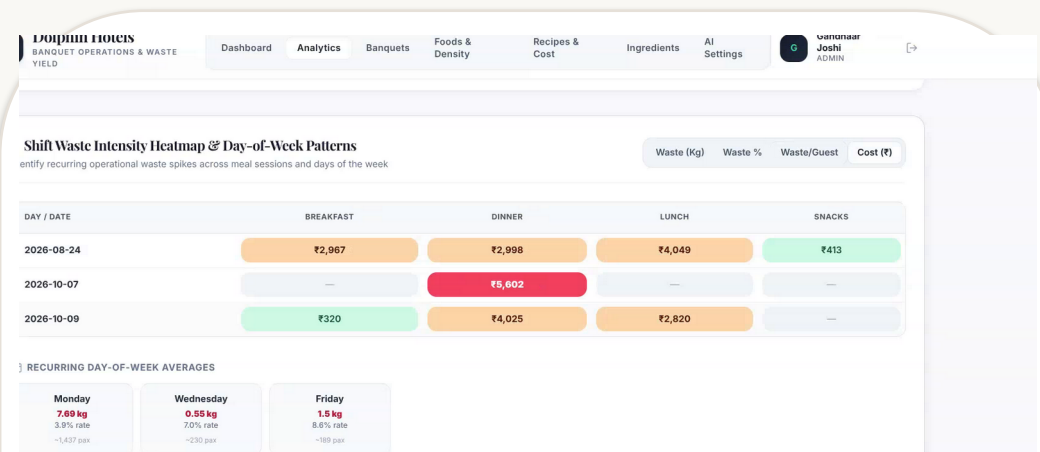
TEMPORAL CLUSTERING

Heatmap Shift Intensity

- Visual Shift Intensity: Color gradient maps high-loss hotspots across Date × Meal Session (Waste kg, %, Cost ₹).
- Day-of-Week Recurring Patterns: Immediately highlights structural variances—such as heavy Sunday night banquet spikes versus compliant Tuesday lunch services.

Turn a Recorded Quantity Into a Transparent Financial Scenario

Distinguishing verified ingredient cost records from hypothetical reduction scenarios provides finance with a disciplined business case.



Current interface snapshot: Savings Simulator & Financial Impact Engine.

SIMULATOR METHODOLOGY

Scenario Modeling Principles

- Dynamic Target Sensitivity: Interactive slider tests 5% to 50% waste curtailment scenarios in real time.
- ERP Cost Integration: Cost per kg automatically derives from master storeroom ingredient rate contracts.
- Direct EBITDA Impact: 100% of avoided food waste expense flows directly to hotel net margin.

VERIFIED SAHARA BASELINE (24 AUG 2026)

Single-Dish Case Study: Plain Rice

Recorded Source Log (24.08.26):

- Lunch Discarded: 26.2 kg (1,722 PAX)
- Dinner Discarded: 29.2 kg (752 PAX)
- Combined Daily Waste: 55.4 kg
- Source Sheet Rate: ~₹20/kg (raw base)
- Recorded Daily Cost: ₹1,108 / day

Illustrative Fully Loaded Cost (₹40/kg):

- Full Cooked Cost: ₹40/kg (ingredients + labor + gas)
- Daily Baseline Cost: 55.4kg × ₹40 = ₹2,216 / day
- 50% Reduction Target: 27.7kg = ₹1,108 / day saved
- 30-Day Potential Savings: ₹33,240 / month (On Plain Rice alone at a single property!)

WHAT-IF REDUCTION SIMULATOR

Interactive Management Scenarios (Monthly Impact)

- -10% Reduction: ₹6,648 / mo potential savings (Conservative batch re-timing).
- -20% Reduction: ₹13,296 / mo potential savings (Moderate, 2-stage pan replenishments).
- -30% Reduction: ₹19,944 / mo potential savings (Optimized, live sensor-triggered refills).
- Note: Illustrative potential savings based on ₹40/kg fully loaded cost model to guide executive targets.

Reliable Decisions Depend on Consistent Records

Centralized analytics require strict data hygiene, transparent calculation definitions, and unified multi-property controls.

AUDIT INTEGRITY

Human-in-the-Loop Controls

- Photo Evidence: Every hardware scan captures a high-resolution image record for verification and chef dispute resolution.
- Manager Override: Authorized supervisors can review and adjust dish classifications, PAX counts, or waste reasons via secure modal.
- Immutable Log: Every weight transaction is timestamped and secured against unauthorized alteration or retroactive editing.
- Tamper Resistance: Eliminates adversarial gaming and ensures finance can rely on data integrity.

MULTI-PROPERTY ROLLOUT

Dolphin Property Portfolio

- Unified Data Schema: Single dashboard filters effortlessly across Sitara Luxury, Sahara Banquets, Tara, and Eureka.
- Cross-Kitchen Benchmarking: Compares grams-per-guest efficiency across kitchen brigades cooking identical menu specifications.
- Centralized Oversight: Headquarters view for Ramoji Film City executive leadership to monitor portfolio-wide trends.
- Scalable Architecture: Start with a pilot station at Sahara and scale seamlessly to Sitara and Tara.

ROLE-BASED GOVERNANCE

Tailored Stakeholder Views

- Executive / GM: High-level monthly savings, sustainability metrics, and property scorecards.
- F&B Controller / Finance: Recipe variance, ingredient wastage costs, inventory drift, and procurement alignment.
- Executive & Sous Chefs: Shift action cards, batch timing recommendations, and item-level waste rankings.
- Kitchen Stewards: Simple, language-agnostic color-coded station confirmations.

Measure. Investigate. Adjust. Measure Again.

The PlateSight analytics continuous improvement loop for Dolphin Group of Hotels.

THE 4-STAGE PILOT ROADMAP

Structured 60-Day Pilot Milestones

Phase 1: Baseline Recording (Weeks 1-2)

Deploy automated monitoring & ingest daily Excel logs. Capture true, unadjusted operational baseline across lunch and dinner buffets.

Phase 2: Recipe & Cost Mapping (Weeks 3-4)

Map Dolphin Group ingredient master rate sheets into the costing engine to establish authoritative rupee valuations.

Phase 3: Targeted Batch Interventions (Weeks 5-6)

Implement action card recommendations on top 3 high-waste items (Plain Rice, Sambar) with staged pan replenishment.

Phase 4: Financial Validation & Review (Weeks 7-8)

Measure post-intervention waste reduction and audit validated rupee savings with hotel GM and finance leadership.



Platesight

PlateSight Intelligence

Building a smarter, more measurable approach to banquet food waste.

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Presenter: Gandhaar Joshi, Co-Founder

Partner Property: Dolphin Group of Hotels • Ramoji Film City